Statistical Analysis of Economic Growth and Its Determinants in India

Data Source: World Bank World Development Indicators (WDI)

Country: India (Code: IND)

Analysis Period: 1991–2025 (35 annual observations)

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Executive Summary & Key Findings

• Over the period 1991–2025 (35 observations), India's average annual GDP Growth was 6.14% (range: -5.78% to 9.69%).

• Multiple OLS regression accounts for 28.37% of the variance in GDP Growth (R2 = 0.2837, F = 2.2969, p = 0.0710).

• Statistically significant predictors at 5% level: FDI*Net*Inflows (coef = -1.8629, p = 0.0492), Gross*Capital*Formation (coef = 0.3362, p = 0.0403).

• Diagnostic Checks: Durbin-Watson statistic (1.74) indicates no severe autocorrelation, while Breusch-Pagan test (p = 0.1222) confirms homoscedasticity.

1. Descriptive Statistics

Analysis of annual macroeconomic indicators for India over the period 1991–2025 (35 observations):

Variable	Count	Mean	Median	Std Dev	Min	Max	Range	Q1	Q3	IQR	Skewness	Kurtosis
GDP_Growth	35	6.14	7.10	2.80	-5.78	9.69	15.47	5.03	7.86	2.83	-2.49	8.85
Inflation	35	6.89	6.33	3.11	2.40	13.87	11.47	4.27	9.23	4.96	0.69	-0.60
FDI*Net*Inflows	35	1.26	1.06	0.80	0.03	3.62	3.59	0.67	1.74	1.07	0.75	0.74
Exports	35	17.64	19.16	5.41	8.49	25.43	16.94	12.01	22.16	10.16	-0.36	-1.35
Gross*Capital*Formation	35	32.10	32.12	5.17	22.72	41.95	19.23	29.04	34.85	5.81	0.00	-0.80
Unemployment	35	7.18	7.61	1.07	4.17	7.86	3.69	7.56	7.64	0.08	-2.30	3.91

2. Probability & Distribution Analysis

Evaluation of distribution shape and formal tests for normality:

Variable	Shapiro-Wilk Stat	Shapiro-Wilk p	Anderson-Darling Stat	Skewness Type	Kurtosis Type	Recommendation
GDP_Growth	0.7735	0.0000	1.9705	Negatively Skewed (Left-tailed)	Leptokurtic (Heavy-tailed/Peaked)	Non-normal distribution (use rank/robust methods)
Inflation	0.9196	0.0139	1.0397	Positively Skewed (Right-tailed)	Platykurtic (Light-tailed/Flat)	Non-normal distribution (use rank/robust methods)

Variable	Shapiro-Wilk Stat	Shapiro-Wilk p	Anderson-Darling Stat	Skewness Type	Kurtosis Type	Recommendation
FDI*Net*Inflows	0.9545	0.1554	0.4367	Positively Skewed (Right-tailed)	Leptokurtic (Heavy-tailed/Peaked)	Normal distribution
Exports	0.9034	0.0049	1.2682	Symmetric	Platykurtic (Light-tailed/Flat)	Non-normal distribution (use rank/robust methods)
Gross*Capital*Formation	0.9691	0.4193	0.3246	Symmetric	Platykurtic (Light-tailed/Flat)	Normal distribution
Unemployment	0.5024	0.0000	8.2095	Negatively Skewed (Left-tailed)	Leptokurtic (Heavy-tailed/Peaked)	Non-normal distribution (use rank/robust methods)

3. Correlation Analysis

Pairwise correlation analysis across 35 annual observations reveals linear (Pearson) and monotonic (Spearman) relationships among economic growth determinants in India. Associations with GDP Growth: Inflation: r = -0.1412 (p = 0.4184, Negative Very Weak / None); FDI*Net*Inflows: r = -0.1189 (p = 0.4962, Negative Very Weak / None); Exports: r = 0.1930 (p = 0.2665, Positive Very Weak / None); Gross*Capital*Formation: r = 0.3180 (p = 0.0626, Positive Weak); Unemployment: r = -0.2095 (p = 0.2270, Negative Weak).

Pairwise Relationships Table

Variable 1	Variable 2	Pearson r	Pearson p	Spearman rho	Spearman p	N Obs	Relationship	Significant (5%)
GDP_Growth	**Inflation**	-0.1412	0.4184	-0.1331	0.4461	35	Negative Very Weak / None	No
GDP_Growth	**FDI*Net*Inflows**	-0.1189	0.4962	0.1090	0.5332	35	Negative Very Weak / None	No
GDP_Growth	**Exports**	0.1930	0.2665	0.2308	0.1822	35	Positive Very Weak / None	No
GDP_Growth	**Gross*Capital*Formation**	0.3180	0.0626	0.4305	0.0098	35	Positive Weak	No
GDP_Growth	**Unemployment**	-0.2095	0.2270	-0.3751	0.0264	35	Negative Weak	No
Inflation	**FDI*Net*Inflows**	-0.0940	0.5911	-0.0950	0.5874	35	Negative Very Weak / None	No
Inflation	**Exports**	-0.2212	0.2016	-0.0401	0.8193	35	Negative Weak	No
Inflation	**Gross*Capital*Formation**	-0.0957	0.5844	-0.0717	0.6823	35	Negative Very Weak / None	No
Inflation	**Unemployment**	0.2869	0.0947	0.2320	0.1800	35	Positive Weak	No
FDI*Net*Inflows	**Exports**	0.6956	0.0000	0.6787	0.0000	35	Positive Strong	Yes
FDI*Net*Inflows	**Gross*Capital*Formation**	0.6248	0.0001	0.6118	0.0001	35	Positive Strong	Yes
FDI*Net*Inflows	**Unemployment**	0.1113	0.5246	0.1182	0.4988	35	Positive Very Weak / None	No
Exports	**Gross*Capital*Formation**	0.8249	0.0000	0.8255	0.0000	35	Positive Very Strong	Yes
Exports	**Unemployment**	-0.3506	0.0389	-0.0826	0.6369	35	Negative Weak	Yes
Gross*Capital*Formation	**Unemployment**	-0.1558	0.3713	-0.1399	0.4227	35	Negative Very Weak / None	No

4. Multiple Linear Regression Model (OLS)

Dependent Variable: `GDP_Growth`

Independent Variables: `Inflation, FDINetInflows, Exports, GrossCapitalFormation, Unemployment`

- **R-squared:** 0.2837
- **Adjusted R-squared:** 0.1602
- **F-statistic:** 2.2969 (p-value = 0.0710)
- **Residual Standard Error:** 2.5695
- **Sample Size (N):** 35 complete observations

Regression Coefficients Table

Variable	Coefficient	Std Error	t-statistic	p-value	95% CI Lower	95% CI Upper	Significant (5%)
const	-1.468261	5.195162	-0.2826	0.7795	-12.0936	9.1570	No
Inflation	-0.111795	0.150015	-0.7452	0.4621	-0.4186	0.1950	No
FDINetInflows	-1.862890	0.907512	-2.0527	0.0492	-3.7190	-0.0068	Yes
Exports	0.009283	0.194612	0.0477	0.9623	-0.3887	0.4073	No
GrossCapitalFormation	0.336227	0.156632	2.1466	0.0403	0.0159	0.6566	Yes
Unemployment	-0.031613	0.538356	-0.0587	0.9536	-1.1327	1.0694	No

5. Regression Diagnostics

Econometric verification of OLS assumptions:

Diagnostic Tests Summary

Test	Statistic	p-value	Passed (5%)	Conclusion
Durbin-Watson Autocorrelation	1.7440	N/A	Yes	No significant autocorrelation (near 2.0)
Breusch-Pagan Heteroscedasticity	8.6870	0.1222	Yes	Homoscedasticity satisfied (Constant Error Variance)
Shapiro-Wilk Residual Normality	0.9001	0.0040	No	Residuals depart from normal distribution

Variance Inflation Factors (VIF)

Predictor Variable	VIF	Multicollinear Warning	Status
Inflation	1.1243	No	Low Multicollinearity (VIF <= 5)
FDINetInflows	2.6905	No	Low Multicollinearity (VIF <= 5)
Exports	5.7055	Yes	Moderate Multicollinearity (VIF > 5)
GrossCapitalFormation	3.3786	No	Low Multicollinearity (VIF <= 5)
Unemployment	1.7242	No	Low Multicollinearity (VIF <= 5)

6. Discussion & Methodological Cautions

Caution: *Association vs. Causation: Neither Pearson/Spearman correlation coefficients nor Ordinary Least Squares (OLS) regression parameters demonstrate causal mechanisms. Observed relationships indicate statistical association only.*

Caution: *Sample Size & Observational Data: Analysis is based on 35 annual time-series observations (1991–2025) for India. Time-series data can be subject to structural breaks, macro shocks (e.g., global recessions), and non-stationarity.*

Caution: *Model Specification & Omitted Variables: OLS regression results reflect the 5 specified indicators (Inflation, FDI Net Inflows, Exports, Gross Capital Formation, Unemployment). Omitted macroeconomic variables (e.g., fiscal policy, monetary stance, institutional changes) may influence the estimated coefficients.*